

Granular Digital Right Manager: A Zero-Trust Document Security and Anti-Leak Platform for Digital Rights Management

Abstract

In today's digital environment, sharing confidential documents carries serious risks of unauthorized copying, leakage, and redistribution. Traditional security solutions, such as basic password protection and standard PDF permissions, are often ineffective because they can be easily bypassed using third-party decryption utilities, virtual PDF printers, or simple copy-paste operations. This project introduces the **Granular Digital Right Manager (GDRM Ecosystem)**, a zero-trust document security platform designed to protect sensitive electronic documents across their entire lifecycle. The system works by encapsulating standard PDF files into encrypted, tamper-evident .gdrm container files. Access to each container is strictly locked to a verified recipient username registered in a cloud backend using Firebase Authentication and Cloud Firestore. When an authorized user opens the file, the application also binds the document to the primary hardware signature of that device, preventing the file from being opened if it is forwarded to an unauthorized computer or phone.

During document viewing, the built-in reader enforces strict client-side protections by disabling text selection, blocking operating system clipboard shortcuts (such as Ctrl+C), and preventing unauthorized screen capture. To secure physical paper workflows, the platform incorporates a dedicated print spooler filter that blocks virtual 'Print to PDF' drivers and only permits physical printing with dynamic, traceable forensic watermarks. Additionally, senders can set timed access rules, including scheduled opening dates, self-destruct countdown timers (ChronoLock), and maximum view limits that destroy decrypted local data upon expiration. By combining cryptographic packaging, cloud identity verification, device hardware locking, and anti-leak reading controls, the GDRM Ecosystem provides an effective, lightweight, and practical security framework for legal, corporate, academic, and government document exchange.

Keywords: Granular Digital Right Manager, GDRM, Document Security, Zero-Trust Architecture, Cryptographic Containers, Hardware Binding, Anti-Copy Protection, Forensic Watermarking, Firebase Authentication.